

Formula sheet

$$\begin{aligned}
 P &= \rho R T_v & \Phi^* &= \Phi_G - \Phi_H - \Phi_E & \frac{\partial P}{\partial z} &= -\rho g & \Delta z &\approx \frac{R}{g} \bar{T}_v \ln \left(\frac{P_1}{P_2} \right) \\
 E_{\text{OLR}} &= \epsilon \sigma T^4 & \sigma &= 5.67 \times 10^{-8} \text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4} \\
 \lambda_{\text{max}} &= \frac{C}{T} & C &= 2897 \mu\text{m} \cdot \text{K} = 2.897 \times 10^{-3} \text{m} \cdot \text{K} \\
 g &= 9.8 \text{m} \cdot \text{s}^{-2} & R &= 287 \text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1} & S &= 1367 \text{W} \cdot \text{m}^{-2} & \rho_0(T=288\text{K}, P=1013\text{hPa}) &= 1.22 \text{kg} \cdot \text{m}^{-3} \\
 P &= \frac{F_{\perp}}{A} \approx \frac{mg}{A} & \rho &= \frac{m}{V} & \Gamma &= -\frac{\partial T}{\partial z} & P_2 &= P_1 \left(\frac{T_2}{T_1} \right)^b & b &= \frac{g}{\Gamma R} \\
 e_s &= e_0 \exp \left(\frac{L_v}{R_v} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right); & T_0 &= 0^{\circ}\text{C} = 273.15\text{K} & e_0 &= e_s(T=0^{\circ}\text{C}) = 611.3\text{Pa} \\
 L_v &= 2.50 \times 10^6 \text{J} \cdot \text{kg}^{-1} & R_v &= 461 \text{J} \cdot \text{K}^{-1} \cdot \text{kg}^{-1}, & C_{p,d} &= 1004 \text{J} \cdot \text{kg}^{-1} \text{K}^{-1} \\
 \Delta E_d &= g \Delta Z + C_{p,d} \Delta T & \Delta E_s &= g \Delta Z + C_{p,d} \Delta T + L_v \Delta q \\
 q &= \frac{(\eta)(e)}{P - (e)(1 - \eta)} = \frac{X}{1 + X} & e &= \frac{(X)(P)}{\eta + X} & X &= \frac{(\eta)(e)}{P - e} & \eta &= \frac{M_{\text{H}_2\text{O}}}{M_{\text{air}}} = 0.622 \\
 T_v &= T \left(\frac{1 + \frac{X}{\eta}}{1 + X} \right) \approx T (1 + 0.61X) & \text{RH} &= 100 \frac{e}{e_s} \approx 100 \frac{q}{q_s} \approx \frac{100X}{X_s} \text{ in } \% \\
 \Theta &= T + \Gamma_d Z = T \left(\frac{P_0}{P} \right)^{\frac{R}{C_{p,d}}} & \frac{R}{C_{p,d}} &= 0.286; & P_0 &= 1000 \text{hPa} \\
 \text{CAPE} &= g \int_{Z_{\text{LFC}}}^{Z_{\text{EL}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} dZ \approx g \sum_{Z_{\text{LFC}}}^{Z_{\text{EL}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} \Delta Z & w_{\text{max}} &= \sqrt{2(\text{CAPE})} \\
 \text{CIN} &= g \int_{Z_0}^{Z_{\text{LFC}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} dZ \approx g \sum_{Z_0}^{Z_{\text{LFC}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} \Delta Z \\
 \Theta_{\text{eq}} &\approx \Theta + \frac{(L_v)(\Theta)(X)}{(C_{p,d})(T)} & Z_{\text{LCL}} &\approx 125(T - T_d) \text{ in metres}
 \end{aligned}$$

here: $\exp(x) \equiv e^x$; Unless otherwise stated, assume $\Gamma_d = 0.010 \text{K} \cdot \text{m}^{-1}$ and $\Gamma_s = 0.006 \text{K} \cdot \text{m}^{-1}$

For dynamics:

$$\begin{aligned}
 \psi &= \begin{cases} 90^{\circ} - \arctan \left(\frac{v}{u} \right) & u < 0 \\ 270^{\circ} - \arctan \left(\frac{v}{u} \right) & u > 0 \end{cases} & |\vec{V}| &= \sqrt{u^2 + v^2} & u &= -|\vec{V}| \sin(\psi) & v &= -|\vec{V}| \cos(\psi) \\
 |\vec{V}_g| &= \frac{1}{|f|\rho} \left| \frac{\partial P}{\partial n} \right| = \frac{g}{|f|} \left| \frac{\partial Z}{\partial n} \right| & u_g &= -\frac{1}{f\rho} \frac{\partial P}{\partial y} = -\frac{g}{f} \frac{\partial Z}{\partial y} & v_g &= \frac{1}{f\rho} \frac{\partial P}{\partial x} = \frac{g}{f} \frac{\partial Z}{\partial x} \\
 |\vec{V}_t| &= \frac{R}{|f|} \left| \frac{dT}{dn} \right| \left(\ln \left(\frac{P_1}{P_2} \right) \right) & v_t &= \frac{R}{f} \frac{\partial T}{\partial x} \left(\ln \left(\frac{P_1}{P_2} \right) \right) & u_t &= -\frac{R}{f} \frac{\partial T}{\partial y} \left(\ln \left(\frac{P_1}{P_2} \right) \right) \\
 \frac{\zeta + f}{D} &= \text{constant} & f &= 2\Omega \sin \phi & \Omega &= \frac{2\pi}{(24)(60)^2} \text{s}^{-1} & \text{Div} &= \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) & \zeta &= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \\
 \Delta y &= \frac{\pi R_E}{180^{\circ}} \Delta \phi & \Delta x &= \frac{\pi R_E \cos \phi}{180^{\circ}} \Delta \theta & R_E &= 6.4 \times 10^6 \text{m} \\
 c_R &= |\vec{V}_g| - \frac{\lambda^2 \Omega \cos(\phi)}{2\pi^2 R_E} & \lambda_c &= \pi \sqrt{\frac{2|\vec{V}_g| R_E}{\Omega \cos \phi}} & k\lambda &= 2\pi = 360^{\circ}
 \end{aligned}$$